

Towards Designing Tools for Young Children’s Privacy in Photos and Support Family-Centered Conversations About Photo Consent

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Young children often don’t know how to give informed consent to being photographed or having their images shared on social media. We conducted an interview of 10 parents of children aged 4 to 5 to understand their opinions on the capture and sharing of photos and videos of their children. Our findings identify two key problems: parents’ lack of technical control over how others share images of their children and their uncertainty around how to teach young children about expressing photo consent. We present two separate work-in-progress resources that were created to address their needs: one, a novel proof-of-concept family-centered mobile app that demonstrates how children’s faces can be detected and blurred automatically in any photos uploaded to social media, and two, a set of materials (a book and sticker sheets) that can be used by parents to discuss with young children the concept of expressing photo consent and having their images shared with others using mobile apps.

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1 Introduction

In this age of social media, young children, aged 3-8, are particularly vulnerable as many cannot give informed consent to being photographed or having their images shared on social media. With the rise of AI that can be used to create deepfakes, it has become even more important for children to understand the implications of their photos shared with other people, especially on social media. Parents post photos of their children online, without considering the long-term implications [1, 7, 8]. Friends and family members may share pictures of kids without the parents’ knowledge or consent. Unintentionally, strangers may also take photos or videos of children who are bystanders and share their faces on the Internet. These uncoordinated sharing practices create complex inter-generational tensions between parents, children, and extended family regarding privacy boundaries and digital agency. Additionally, current social platforms lack responsible, family-centered governance tools that protect children’s identities while respecting values of shared memory-making. Furthermore, even privacy-conscious parents struggle to navigate these conversations, lacking age-appropriate tools to help young children understand consent. To address these gaps, we conducted an interview study with 10 parents of 4–5-year-olds, and present two work-in-progress solutions: KidSafe, a family-centered mobile

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app that automates face-blurring, and a novel educational storybook and sticker tool to facilitate early parent-child conversations around photo consent.

All existing Privacy Enhancing Technologies (PETs) that protect faces in photos are built for adults. These solutions use visual markers to communicate privacy preferences directly to cameras [3, 5, 6], methods that prevent automatic detection of faces in photos by disrupting facial recognition algorithms [2], as well as Secondary Channel Picture PETs that rely on communication of recording preferences between the person being recorded and the camera [4, 9]. Additionally, feedback from parents in our study show that they needed control over photos containing their children's faces and transparency about the algorithms used, so, the existing PETs will not meet their needs.

2 Pilot Interview with Parents

We conducted a 30 minute pilot semi-structured interview study with 10 parents of 4-5 year old children. We recruited participants by circulating a flyer at the researchers' institutes and through word of mouth. The interviews were conducted by a researcher one-on-one over Zoom. The study was approved by our Institutional Review Boards. The interview was divided into three phases: photo-taking practices and contexts, parent-child dynamics and consent, sharing and protective technology preferences.

2.1 Key Findings

We audio recorded the Zoom meetings and conducted thematic analysis on the meeting transcripts using Braun and Clarke's framework. First, three researchers independently created initial open codes. The team then met collectively to compare codes and collaboratively clustered the codes into overarching themes.

2.1.1 Views on Social Media Posts with Childrens' Faces. All ten parents were hesitant to post photos of their children's faces on social media or the internet. Several participants used private accounts restricted to close friends and family, while others preferred direct services like Apple Shared Albums. Parents were primarily concerned about losing control over their children's images. P1167 said, "people can just... screenshot it, and then the whole world has access to that picture." Furthermore, participants expressed fear that third parties could misuse their children's images, specifically raising concerns about deepfakes and AI tools being used to generate inappropriate content. P1167 added "Photoshopping them, putting them into, you know, AI ChatGPT, creating inappropriate pictures. And, you know, that's the stuff that I'm afraid of, that I don't want images of my child to fall into the wrong hands"

2.1.2 Emoticons and face blurring on children's faces. All ten parents were familiar with the use of emoticons or face blurring to protect children's identity and were open to using these techniques. Parents saw value for face blurring or obstruction, especially in group photos involving children, such as birthday parties, where the photo-taker does not have explicit consent from other families. In these public contexts, participants expressed a strong willingness to use automated tools to instantly obscure other children's faces before sharing images. Specially, multiple parents saw the value in adding this in group photos and public settings. P1482 said "Like if maybe they were at a birthday party or something and the parent wanted to share a photo but they weren't sure if I would approve and they put an emoji over it (my child's face), I guess I don't think I would be upset about that." If it was a stranger taking the photo, for example, a child-centered business owner, they would like their child's face to be blocked by default.

However, multiple participants expressed concerns for data privacy when using any solution that used AI algorithms, specifically face detection and recognition algorithms. Their concerns were about data use and storage, specifically whether the app would itself store the unblurred photo on a server and use it for training the algorithms.

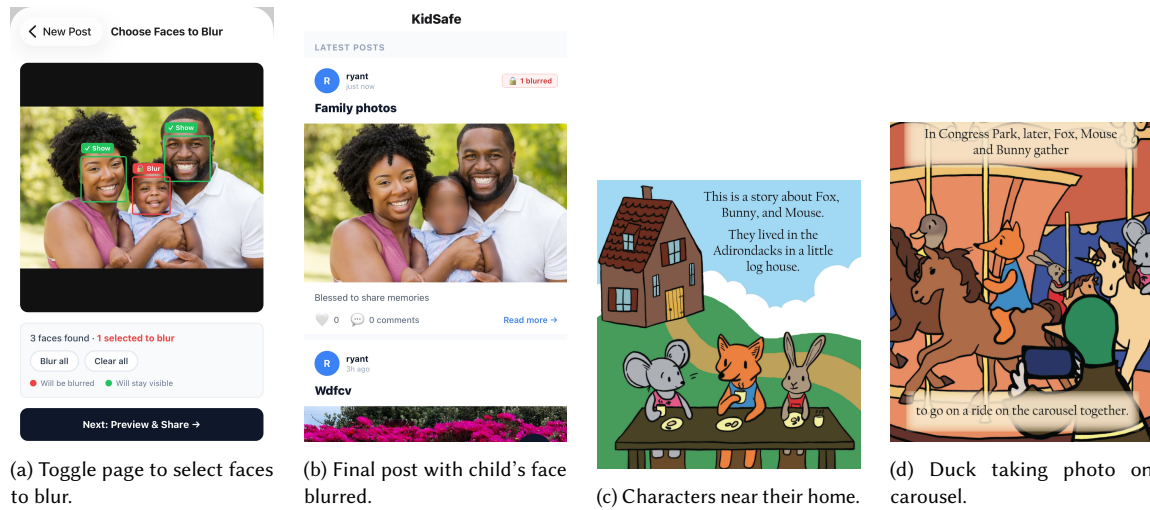


Fig. 1. Screenshots of the KidSafe app (a, b) and the educational picture book (c, d).

P1453 said “I think just right off the bat, just thinking about it, it’s probably going to be a technology that uses AI. And AI is a platform that is completely unregulated with no safety precautions usually built in, at least like big LLMs like ChatGPT. If they build it from there, it’s really hard to do security with something like AI. And most of the programs that I know that do things like this are explicitly using those technologies. So my concern or like my guess or worry would be that it would put an emoji over your child’s face and then there would be no backend security whatsoever. So I would I like would at least read what they did to create the software before I used it.”

Parents overwhelmingly demanded final agency. P1167 said “My preference would be that if somebody wants to take a picture of my child, that they speak to me about it first. And I wonder if technology like that would just make it easier for people to feel like they can do those things without getting consent.”

2.1.3 Parent-Child Dynamics While Capturing Photos. Some children were capable of expressing consent to being in photographs as young as three, according to the parents. Some times, the kids themselves initiate the photographs. P1453 said, “He’ll say, mommy, can you take a picture of me and send it to grandma?” Some children say no either by explicitly saying so, or by running away. Some even ask their parents to delete the photo if they are not happy with it.

Parents were unsure of how to bring the topic up with their children. P1414 said, “I haven’t had the conversation with him, but I have definitely thought about it because there are some kiddos in his preschool class that like they don’t allow the school to take pictures and share them. And I respect that, but also I think it would be a good conversation to have because I want him to be able to like feel comfortable saying, if someone else were to take a picture, I would want him to feel comfortable saying like, if he didn’t want to be included, because I wouldn’t want someone else to post like a picture of my kid online.” P1453 said “I haven’t had any conversations as of yet because I think I’m still trying to figure out how to phrase it so that a four-year-old can kind of understand the gist of what I’m saying.”

3 Automatic Face Blurring: KidSafe App

Our interview findings revealed that all participating parents expressed significant privacy concerns regarding unauthorized social media posts of their children. To address this, we designed KidSafe, a mobile application for photo-sharing

that protects children’s privacy. We developed this app to serve as a proof-of-concept of a social media app which prevents users from uploading images containing a child’s face. Before a photo is posted, the application detects every face in it, estimates each person’s age, and automatically marks children’s faces to be blurred. The design is meant to be ‘safe-by-default’, the children’s faces, that is, any faces that the app recognizes as younger than eighteen, start in the “blur” state; the user must take a deliberate action to reveal a child’s face. Only verified parent’s accounts can unblur children’s faces, while the adults can be blurred or unblurred by anyone. The verification is not implemented yet, but will be in the future.

The user reviews the photo, taps any face to toggle its blur on or off, and the final image is blurred permanently before it is uploaded and shared. The blur is baked into the image pixels before upload so only the blurred version is ever stored or shared. The original unblurred photo never reaches the feed.

4 Book, Sticker Sheets and a Script to Support Conversation

To address interview participants’ difficulties in initiating conversations about photo consent with young children, we designed an 8-page picture book for parents and educators. The picture book shows three wild animals Fox, Bunny, and Mouse visiting a local town. The story contrasts familiar, safe spaces (e.g., home) with crowded public settings involving friends and strangers.

In addition to the book, we also created two sticker sheets with two scenes from the book to allow the child to come up with their own story of what photo is being captured and by whom, who is in the photo and how are they feeling (Refer 1). We also provided a script that parents could use to discuss the topic of consent to being photographed. Together, these playful, child-centered tools serve as a shared medium for parents and educators to scaffold early conversations, empowering young children to recognize, articulate, and establish their privacy boundaries.

So far, we have pilot tested with only one parent with 4- and 8-year-old girls. The 4 year old enjoyed the book and the stickers but was unable to talk about whether the characters in the book wanted to be photographed; she was able to talk about her own experiences, what kind of photos she wanted to be in and who these photos could be shared with. The 8 year old was able to talk about the characters and how they would feel about the, for example, she thought Mama Duck should not take the video of her duckling on the carousel especially if she could avoid it, since Fox, Bunny, and Mouse could be captured as well.

We are in the process of conducting further studies to test whether our materials can be used to have meaningful conversations with young children (ages 3-8) about photo consent and their photos being shared with others.

5 Discussion and Conclusion

We presented two work-in-progress solutions: 1. KidSafe, a proof-of-concept app for automated face-blurring that addresses parents’ need to protect children’s privacy, as identified in our study and 2. materials that parents and educators can use to discuss photo consent and having images shared on social media.

These developments raise broader questions, such as, 1) how can parents facilitate effective discussions with children about what it means to be in a photograph 2) how can photos with kids’ faces be shared with other people on the Internet safely, and how can parents have conversations with extended family and friends about their permissions and boundaries 3) what are the implications of using AI algorithms in family photography 4) how can the research community design futures for family-centered AI transparently, such that families can stay informed about data storage and usage for training, and how their data is being used by other consumers.

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